

Assignment No.-2

Q.1. Define Student t-distribution?

Ans. Suppose, U & V are independent random variable if U follows $N(0, 1)$ and V follows chi-square distribution with n degrees of freedom.

Then, $t = \frac{U}{\sqrt{V/n}}$ is said to follow t-distribution with parameter n which is called as n degrees of freedom.

$$\text{i.e., } t = \frac{N(0, 1)}{\sqrt{\frac{\chi^2}{n}}}$$

Q.2. Show that odd central moments of the t-distribution are zero?

Ans.

$$\therefore \mu_{2r+1} = E(T^{2r+1})$$

$$= \int_{-\infty}^{\infty} t^{2r+1} f(t) dt$$

$$\therefore \mu_{2r+1} = \int_{-\infty}^{\infty} \frac{1}{\sqrt{n} \beta\left(\frac{1}{2}, \frac{n}{2}\right)} \cdot \frac{t^{2r+1}}{\left(1 + \frac{t^2}{n}\right)^{\frac{n+1}{2}}} dt$$

$\therefore$ Since, the integrand is odd function is zero.

$$\therefore \mu_{2r+1} = 0 \quad ; \quad \text{for } r = 1, 2, 3, \dots$$

Thus, all the moments of odd order are zero.

Therefore, first four central moments are,

$$\mu_1 = 0,$$

$$\mu_2 = \frac{n}{n-2},$$

$$\mu_3 = 0,$$

$$\mu_4 = \frac{3n^2}{(n-2)(n-4)}$$

Coefficients of skewness $\beta_1 = 0$, $\gamma_1 = 0$.

Hence, the distⁿ symmetric about mean (zero).

Also, the coefficient of kurtosis.

$$\beta_1 = \frac{\mu_4}{\mu_2^2}$$

$$= \frac{3n^2}{(n-2)(n-4)} \cdot \frac{(n-2)^2}{n^2}$$

$$= \frac{3(n-2)}{n-4}$$

$$= \frac{3(n-4+2)}{n-4}$$

$$\therefore \beta_1 = 3 + \frac{6}{n-4} > 3$$

$$\therefore \gamma_2 = \beta_1 - 3$$

$$= 3 + \frac{6}{n-4} - 3$$

$$\therefore \gamma_2 = \frac{6}{n-4}$$

$\therefore$ Hence, the distribution is leptokurtosis.

Q.3. Derive the second central moment of the t-distribution?

Ans.

$$\therefore \mu_{2v} = n \cdot \frac{\beta\left(\frac{3}{2}, \frac{n-1}{2}\right)}{\beta\left(\frac{1}{2}, \frac{n}{2}\right)}$$

$$= n \frac{\sqrt{\frac{3}{2}} \sqrt{\frac{n-1}{2}}}{\sqrt{\frac{1}{2}} \sqrt{\frac{n}{2}}}$$

$$= n \cdot \frac{\frac{1}{2} \sqrt{\pi} \sqrt{\frac{n-1}{2}}}{\sqrt{\pi} \cdot \left(\frac{n-1}{2}\right) \sqrt{\frac{n}{2}}}$$

$$= \frac{n}{2}$$

$$\mu_2 = \frac{n}{n-2}$$

Q.4. If x follow t-distribution with r d.f. then $\text{Var}(x)$ is?

Ans. Given,

$$n = r.$$

$$\text{Var}(X) = ?$$

$$\therefore \text{Var}(X) = \frac{n}{n-2}$$

$$= \frac{5}{5-2}$$

$$= \frac{5}{3}$$

$$\therefore \text{Var}(X) = 1.66$$

Q. 5. If X is a random variable having t -distribution having t -distribution with n d.f. such that $\frac{\mu_6}{\mu_4} = 7$ then n is equal to ? then n is equal to ?

Ans. Given,

$$\frac{\mu_6}{\mu_4} = 7, n = ?$$

$$\therefore \mu_4 = \frac{3n^2}{(n-2)(n-4)} \quad \text{for } n > 4$$

$$\mu_6 = \frac{15n^3}{(n-2)(n-4)(n-6)} \quad \text{for } n > 6$$

$$\therefore \frac{\mu_6}{\mu_4} = \frac{15n^3}{\frac{3n^2}{(n-2)(n-4)}} = \frac{15n^3 \cdot (n-2)(n-4)}{3n^2}$$

$$= \frac{5n^3}{3n^2(n-6)}$$

$$\therefore \frac{\mu_6}{\mu_4} = \frac{5n}{n-6}$$

$$\therefore \frac{5n}{n-6} = 7$$

$$\therefore 5n = 7(n-6)$$

$$\therefore 5n = 7n - 42$$

$$\therefore -2n = -42$$

$$\therefore 7n - 5n = 42$$

$$\therefore 2n = 42$$

$$\therefore n = \frac{42}{2}$$

$$\boxed{n = 21}$$

Q. 6. If t follow student's t -distribution with 15 d.f. find

i) C such that $P(-c \leq t \leq c) = 0.8$?

ii) k such that $P(t^2 \geq k) = 0.05$?

iii) A such that $P(t \leq A) = 0.1$?

Ans. Let,

$$i) P(-c \leq t_{15} \leq c) = 0.8$$

$$\therefore 1 - 2 \cdot P(-c > t_{15} > c) = 0.8$$

$$\therefore -2 \cdot P(-c > t_{15} > c) = 0.8 - 1$$

$$\therefore -2 \cdot P(-c > t_{15} > c) = -0.2$$

$$\therefore P(-c > t_{15} > c) = \frac{-0.2}{-2}$$

$$\therefore P(-c > t_{15} > c) = 0.1$$

$$\therefore P(-c \leq t_{15} \leq c) = 1.341$$

$$ii) P(t^2 \geq k) = 0.05?$$

→

$$\therefore P(|t_{15}| \geq \sqrt{k}) = 0.05$$

From statistical table we get,

$$P(|t_{15}| > 2.131) = 0.05,$$

$$\text{Therefore, } \sqrt{k} = 2.131$$

$$\therefore P(t^2 \geq k) = 0.5412$$

$$iii) P(t \leq a) = 0.1?$$

→

$$\therefore P(t \leq a) = 0.1 \quad a < 0$$

$$\therefore P(t \geq -a) = 0.1 \text{ Due to symmetry.}$$

$$\therefore P(|t| \geq -a) = 0.2$$

From statistical table,

$$\therefore P(|t_{15}| > 1.341) = 0.2$$

$$\therefore a = -1.341$$

$$\therefore P(t \leq a) = 0.1$$

Q.7. IF t_n follow Student's t -distribution with n d.f. find
i) $P(t_{30} < 0.531)$
ii) $P(t_{30} > 1.697)$

$$PPP) P(|t_{30}| > 1.812)$$

$$iv) P(|t_{30}| \leq 2.305)?$$

Ans. Let,

$$i) P(t_{30} < 0.531)?$$

→

$$\therefore 1 - P(t_{30} < 0.531)$$

$$\therefore 1 - 0.30$$

$$\therefore 0.70$$

$$ii) P(t_{30} > 1.697)?$$

→

$$\therefore 0.05$$

$$iii) P(|t_{30}| > 1.812)?$$

→

$$\therefore 2 \cdot P(t_{30} > 1.812)$$

$$\therefore 2 \cdot (0.05)$$

$$\therefore 0.10$$

$$iv) P(|t_{30}| \leq 2.305)?$$

→

$$\therefore 1 - 2 \cdot P(t_{30} > 2.305)$$

$$\therefore 1 - 2 \cdot (0.025)$$

$$\therefore 1 - 0.05$$

$$\therefore 0.95$$

Q. 1. Define F-distribution with n_1 & n_2 d.f. and derive its P.d.f.?

Ans. Suppose, U & V are independent χ^2 random variables with n_1 & n_2 degrees of freedom respectively then, $F = \frac{U/n_1}{V/n_2}$ is said to follow F-distribution with parameter n_1 & n_2 .

$\therefore$ P.d.f :-

i) Let, $F = \frac{U/n_1}{V/n_2}$ and $z = V$ clearly $0 < z < \infty$
 $0 < f < \infty$

ii) We get,

$$\therefore F = \frac{U/n_1}{V/n_2}$$

$$\therefore F = \frac{U}{n_1} \times \frac{n_2}{V}$$

$$\therefore F = \frac{U n_2}{n_1 V}$$

$$\therefore U = F \times V \times \frac{n_1}{n_2}$$

$\therefore$ Put $z = V$

$$\therefore U = F \times z \times \frac{n_1}{n_2}$$

$$\text{i.e., } U = \left(\frac{n_1}{n_2}\right) F \cdot z \text{ \& } V = z$$

By using Jacobian transformation,

$$\therefore J = \frac{\partial(U, V)}{\partial(F, z)}$$

$$= \begin{vmatrix} \frac{\partial U}{\partial F} & \frac{\partial U}{\partial z} \\ \frac{\partial V}{\partial F} & \frac{\partial V}{\partial z} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{n_1}{n_2} \cdot z & \frac{n_1}{n_2} \cdot F \\ 0 & 1 \end{vmatrix}$$

$$= \left(\frac{n_1}{n_2} \cdot z\right) \times 1 - \left(\frac{n_1}{n_2} \cdot F\right) \times 0$$

$$= \left(\frac{n_1}{n_2} \cdot z\right) - 0$$

$$\therefore J = \left(\frac{n_1}{n_2}\right) \cdot z$$

Q. 2. Derive raw moment and find mean and variance of the distribution?

Ans.

$$\therefore \mu_r' = \left(\frac{n_2}{n_1}\right)^r \cdot \frac{\beta\left(\frac{n_1}{2} + r, \frac{n_2}{2} - r\right)}{\beta\left(\frac{n_1}{2}, \frac{n_2}{2}\right)}$$

Put $r = 1$

$$\therefore \mu_1' = \text{mean} = \frac{n_2}{n_2 - 2}$$

Put $r = 2$

$$\therefore \mu_2' = \left(\frac{n_2}{n_1}\right)^2 \cdot \frac{\beta\left(\frac{n_1}{2} + 2, \frac{n_2}{2} - 2\right)}{\beta\left(\frac{n_1}{2}, \frac{n_2}{2}\right)}$$

$$\therefore \mu_2' = \left(\frac{n_2}{n_1}\right)^2 \cdot \left(\frac{\sqrt{\frac{n_1}{2} + 2} \cdot \sqrt{\frac{n_2}{2} - 2}}{\sqrt{\frac{n_1}{2}} \cdot \sqrt{\frac{n_2}{2}}}\right)$$

$$\therefore u_2' = \left(\frac{n_2}{n_1}\right)^2 \cdot \frac{\left[\frac{n_1+2}{2}\right] \cdot \left[\frac{n_2-2}{2}\right]}{\left[\frac{n_1+x}{2} + \frac{n_2-x}{2}\right]} \cdot \frac{\left[\frac{n_1}{2}\right] \left[\frac{n_2}{2}\right]}{\left[\frac{n_1+n_2}{2}\right]}$$

$$= \left(\frac{n_2}{n_1}\right)^2 \cdot \frac{\left[\frac{n_1+2}{2}\right] \cdot \left[\frac{n_2-2}{2}\right]}{\left[\frac{n_1+n_2}{2}\right]} \times \frac{\left[\frac{n_1}{2}\right] \left[\frac{n_2}{2}\right]}{\left[\frac{n_1+n_2}{2}\right]}$$

$$= \left(\frac{n_2}{n_1}\right)^2 \cdot \frac{\left[\frac{n_1+2}{2}\right] \cdot \left[\frac{n_2-2}{2}\right]}{\left[\frac{n_1}{2}\right] \left[\frac{n_2}{2}\right]}$$

$$= \left(\frac{n_2}{n_1}\right)^2 \cdot \frac{\left[\frac{n_1+2}{2}-1\right] \left(\frac{n_1+2}{2}\right) \cdot \left[\frac{n_2-2}{2}\right]}{\left[\frac{n_1}{2}\right] \left[\frac{n_2-1}{2}\right] \left(\frac{n_2-1}{2}\right)}$$

$$= \left(\frac{n_2}{n_1}\right)^2 \cdot \frac{\left[\frac{n_1+1}{2}\right] \left(\frac{n_1+1}{2}\right) \left[\frac{n_2-2}{2}\right]}{\left[\frac{n_1}{2}\right] \left[\frac{n_2-1}{2}\right] \left(\frac{n_2-1}{2}\right) \left(\frac{n_2-1}{2}\right)}$$

$$= \left(\frac{n_2}{n_1}\right)^2 \cdot \frac{\left[\frac{n_1+1}{2}\right] \left(\frac{n_1+1}{2}\right) \left[\frac{n_2-2}{2}\right]}{\left[\frac{n_1}{2}\right] \cdot \left[\frac{n_2-2}{2}\right] \left(\frac{n_2-2}{2}\right) \left(\frac{n_2-1}{2}\right)}$$

$$= \left(\frac{n_2}{n_1}\right)^2 \cdot \frac{\left[\frac{n_1+x}{2}\right] \left(\frac{n_1+x}{2}\right) \left(\frac{n_1+1}{2}\right)}{\left[\frac{n_1}{2}\right] \left(\frac{n_2-2}{2}\right) \left(\frac{n_2-1}{2}\right)}$$

$$= \left(\frac{n_2}{n_1}\right)^2 \cdot \frac{\left(\frac{n_1}{2}\right) \left(\frac{n_1+1}{2}\right)}{\left(\frac{n_2-2}{2}\right) \left(\frac{n_2-1}{2}\right)}$$

$$= \left(\frac{n_2^2}{n_1}\right) \cdot \frac{\left(\frac{n_1+2}{2}\right)}{\left(\frac{n_2-2}{2}\right) \cdot \left(\frac{n_2-2}{2}\right)}$$

$$= \left(\frac{n_2^2}{n_1}\right) \cdot (n_1+2) \cdot (n_2-4) \cdot (n_2-2)$$

$$\therefore u_2' = \frac{n_2^2 (n_1+2)}{n_1 (n_2-4) (n_2-2)}$$

$$\therefore V(x) = u_2' - (u_1')^2$$

$$= \frac{n_2^2}{n_1} \cdot \frac{(n_1+2)}{(n_2-4) (n_2-2)} - \frac{n^2}{(n^2-2)}$$

$$\therefore V(x) = \frac{2n_2^2 (n_1 + n_2 - 2)}{n_1 (n_2 - 2)^2 (n_2 - 4)}$$

Q.3 Derive mode of F-distribution?

Ans. Let,

$$P(F) = \frac{\left(\frac{n_1}{2}\right)^{\frac{n_1}{2}-1} F^{\frac{n_1+n_2}{2}}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right) \left(1 + \frac{n_1}{n_2} F\right)^{\frac{n_1+n_2}{2}}}$$

$$\therefore P(F) = C \cdot \frac{F^{\frac{n_1}{2}-1}}{\left(1 + \frac{n_1}{n_2} F\right)^{\frac{n_1+n_2}{2}}}$$

Where,

$$C = \frac{\left(\frac{n_1}{2}\right)^{\frac{n_1}{2}-1}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)}$$

Taking logarithms on both sides,

$$\therefore \log P(F) = \log C + \left(\frac{n_1}{2} - 1\right) \log F - \left(\frac{n_1+n_2}{2}\right) \log \left(1 + \frac{n_1}{n_2} F\right)$$

Differentiating both the sides w.r.t. F, we get,

$$\therefore \frac{P'(F)}{P(F)} = 0 + \frac{n_1}{2} - 1 - \left(\frac{n_1+n_2}{2}\right) \frac{\frac{n_1}{n_2}}{1 + \frac{n_1}{n_2} F}$$

$$\therefore P'(F) = \left\{ \frac{n_1-2}{2F} - \frac{n_1(n_1+n_2)}{2n_2} \times \frac{n_2}{n_2+n_1F} \right\} P(F) \quad \text{--- (1)}$$

$\therefore P'(F) = 0$, gives,

$$\therefore \frac{n_1-2}{2F} = \frac{n_1(n_1+n_2)}{2(n_2+n_1F)}$$

$$\therefore (n_2+n_1F)(n_1-2) = n_1(n_1+n_2)F$$

$$\therefore n_1(n_1-2)F - n_1n_2F = n_1^2F - n_2(n_1-2)$$

$$\therefore F = \frac{n_2}{n_2+2} \cdot \frac{n_1-2}{n_1}; \text{ if } n_1 > 2$$

Differentiating eqⁿ (1) w.r.t. F we can verify,

$$\therefore P''(F) \leq 0 \text{ at } \frac{n_2(n_1-2)}{n_1(n_2+2)}$$

Hence,

$$\text{The mode is } \left(\frac{n_2}{n_2+2}\right) \left(\frac{n_1-2}{n_1}\right).$$

Q.4 Find quantile of F-distribution?

Ans. Let,

$$Q_1 = P(F_{(n_1, n_2)} \leq Q_1) = \frac{1}{4}$$

$$\therefore P\left(\frac{1}{F_{(n_1, n_2)}} \geq \frac{1}{Q_1}\right) = \frac{1}{4}$$

$$\therefore P\left(F_{(n_1, n_2)} \geq \frac{1}{Q_1}\right) = \frac{1}{4}$$

$$\therefore P\left(F_{(n_1, n_2)} \leq \frac{1}{Q_1}\right) = 1 - \frac{1}{4}$$

$$\therefore Q_1 = P(F_{(n,n)} \leq \frac{1}{Q_1}) = \frac{3}{4} \quad \text{--- (1)}$$

$$\therefore Q_3 = P(F_{(n,n)} \leq Q_3) = \frac{3}{4} \quad \text{--- (2)}$$

$$\therefore \frac{1}{Q_1} = Q_3$$

$$Q_1 \times Q_2 = 1$$

$$\therefore Q_3 = \frac{1}{Q_1}$$

$$\therefore Q_1 = \frac{1}{Q_3}$$

Inter-Relationship Among T , F , χ^2 Variable.

Q.5. Write inter relationship among T , F , χ^2 Variables?

Ans. Let,

i) Relation between T & F

Note that,

If U & V are independent r.v.s. $U \sim N(0,1)$

$V \rightarrow \chi^2_n$ then we get, $t = \frac{U}{\sqrt{V/n}}$ as t .

variate with n d.f. clearly, $t^2 = \frac{U^2}{V/n}$

$$= \frac{\chi^2_1 V_1}{\chi^2_n / n} = F_{(1,n)}$$

$\therefore$ Thus,

We get, if $t \rightarrow t_n$ then $t^2 \rightarrow F_{(1,n)}$.

Q.6. Let, $X \sim F(4, 20)$. Find i) $P(X > 3.51)$

ii) $P(X \leq 0.207)$?

Ans. Let,

$$X \sim F(4, 20)$$

$$M_1 = 4,$$

$$M_2 = 20.$$

i) $P(X > 3.51)$?

$\rightarrow$

$$\therefore P(X > 3.51) = 1 - P(X \leq 3.51)$$

$$= 1 - 0.98$$

$$\therefore P(X > 3.51) = 0.025$$

ii) $P(X \leq 0.207)$?

$\rightarrow$

$$\therefore P(X \leq 0.207) = \frac{1}{x}$$

$$= \frac{1}{0.207}$$

$$\therefore P(X \leq 0.207) = 4.83$$

$$\therefore P(F(20, 4) > 4.83)$$

$$M_1 = 20,$$

$$M_2 = 4$$

$\therefore$ At $\alpha = 0.10$, the value is 4.59.

$\therefore$ At $\alpha = 0.05$, the value is 5.80.

$$\therefore P(X \leq 0.207) = 0.0685$$

Q.7 If X follows $F(10, 11)$ d.f. then find mean, median, $Y = \frac{1}{X}$ hence find mean, mode of the distribution Y ?

Ans. Let,

$\therefore$ Properties of X :-

$$n_1 = 10, n_2 = 11$$

$$u_1 = \frac{n_2}{n_2 - 2} = \text{Mean}$$

$$= \frac{11}{11-2}$$

$$= \frac{11}{9}$$

$$\therefore u_1 = 1.222$$

$$\therefore \text{Mean}(u_1) = \frac{n_2}{n_2 - 2}$$

$$= \frac{10}{10-2}$$

$$= \frac{10}{8}$$

$$\therefore \text{Mean} = 1.25$$

$\therefore$ Mode of Y :-

$$\therefore \text{Mode} = \frac{n_1 - 2}{n_1} = \frac{n_2 + 1}{n_2 + 2} \quad (\text{for } n_2 \geq 2)$$

$$= \left(\frac{11-2}{11} \right) \cdot \left(\frac{10}{10+2} \right)$$

$$= \frac{9}{11} \cdot \frac{10}{12}$$

$$= \frac{90}{132}$$

$$\therefore \text{Mode}_x = 0.681$$

$\therefore$ Median (M_y) :-

$$\therefore F(10, 11) \approx 0.9997$$

$\therefore$ Median (M_y) :-

$$\therefore F(11, 10) \approx 1.006$$

$$\therefore \text{Mode}(M_y) = \frac{n_1 - 2}{n_1} \cdot \frac{n_2}{n_2 + 2}$$

$$= \frac{10-2}{10} \cdot \frac{11}{11+2}$$

$$= \frac{8}{10} \cdot \frac{11}{13}$$

$$= \frac{88}{130}$$

$$\therefore M_y = 0.6769$$

Q.8 Tests based on Sampling distributions.

1) Tests based on chi-square distribution :-

Q.1 Explain the test procedure for testing the independence

Ans. two attribute in $r \times s$ contingency table? χ^2 test for independent of two attribute arrange in $r \times s$ contingency table.

- Observed frequency table :-

A \ B	B ₁	B ₂		B _s	Total
A ₁	O ₁₁	O ₁₂		O _{1s}	(A ₁)
A ₂	O ₂₁	O ₂₂		O _{2s}	(A ₂)
⋮	⋮	⋮	⋮	⋮	⋮
A _r	O _{r1}	O _{r2}		O _{rs}	(A _r)
Total	(B ₁)	(B ₂)		(B _s)	N

$$\therefore N = \sum_{i=1}^r \sum_{j=1}^s o_{ij} = \text{Total observed freq.}$$

$$\therefore (A_i) = \sum_{j=1}^s o_{ij} = \text{Total observed freq. in } i^{\text{th}} \text{ row.}$$

$$\therefore (B_j) = \sum_{i=1}^r o_{ij} = \text{Total observed freq. in } j^{\text{th}} \text{ row.}$$

- Expected frequency table :-

A \ B	B ₁	B ₂		B _s	Total
A ₁	e ₁₁	e ₁₂		e _{1s}	
A ₂	e ₂₁	e ₂₂		e _{2s}	
⋮	⋮	⋮	⋮	⋮	⋮
A _r	e _{r1}	e _{r2}		e _{rs}	

$$\text{Total} = \sum \sum e_{ij} = N$$

H₀ : Two attributes are independent.

H₁ : Two attributes are not independent.

$$\therefore e_{ij} = \frac{(A_i)(B_j)}{N}$$

- Test statistic :-

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^s \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

OR

$$\therefore \chi^2 = \sum_{i=1}^r \sum_{j=1}^s \left(\frac{o_{ij}^2}{e_{ij}} \right) - N$$

Q.2. Describe χ^2 test for goodness of fit. state the assumption we make while applying the test?

Ans. For a given data (freq. distribution) we try to fit some probability distribution. which distribution will fit properly may be a question of interest. In such cases we want to test the appropriateness of the fit. Hence, we desire to test H_0 : Fitting of the probability distribution to given data is proper (good). The test based on χ^2 -distribution is used to test this H_0 . It is called χ^2 -test of goodness of fit.

Suppose,

$o_1, o_2, \dots, o_k$ be a set of observed freq. and $e_1, \dots, e_i, \dots, e_k$ be corresponding expected freq. obtained order.

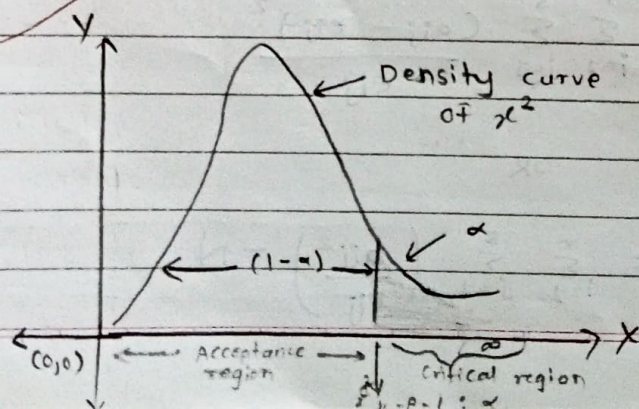
$$\therefore \sum_{i=1}^k o_i = N = \sum_{i=1}^k e_i$$

If H_0 is true, then the statistic

$$\therefore \chi^2 = \sum_{i=1}^k \frac{(o_i - e_i)^2}{e_i} = \sum_{i=1}^k \left(\frac{o_i^2}{e_i} \right) - N$$

$\therefore \chi^2$ -distribution with $(k - p - 1)$ d.f. In this case the region at I.o.s. α is, $\chi^2_{k-p-1} > \chi^2_{k-p-1; \alpha}$.

$\therefore$ It is shown by the shaded region in figure,



Q.3. → Described the test procedure of McNemar's test? We have seen the test of independence of two attributes each at 2 levels (2×2 contingency table). However, this test cannot be used to test the hypothesis of symmetry for 2×2 contingency table. In this case McNemar's test is applied we want to test that the two attributes are equally effective. Suppose the 2×2 contingency table is given by,

A \ B	B	B_1	B_2	Total
	A			
A_1		O_{11}	O_{12}	$O_{1.}$
A_2		O_{21}	O_{22}	$O_{2.}$
Total		$O_{.1}$	$O_{.2}$	N

Where O_{ij} 's are observed frequencies $i, j = 1, 2$. The corresponding table of expected frequencies is given by,

A \ B	B	B_1	B_2	Total
	A			
A_1		e_{11}	e_{12}	$e_{1.}$
A_2		e_{21}	e_{22}	$e_{2.}$
Total		$e_{.1}$	$e_{.2}$	N

Accordingly the table of probabilities is,

A \ B	B	B_1	B_2	Total
	A			
A_1		$\frac{e_{11}}{N} = P_{11}$	$\frac{e_{12}}{N} = P_{12}$	$\frac{e_{1.}}{N} = P_{1.}$
A_2		$\frac{e_{21}}{N} = P_{21}$	$\frac{e_{22}}{N} = P_{22}$	$\frac{e_{2.}}{N} = P_{2.}$
Total		$\frac{e_{.1}}{N} = P_{.1}$	$\frac{e_{.2}}{N} = P_{.2}$	1

In this test our null hypothesis H_0 is of equality of marginal distⁿ against two sided alternative that they are not equal.

In other words,

$$H_0: P_1 = P_2 \text{ against } H_1: P_1 \neq P_2$$

$$\text{or } H_0: P_1 = P_2 \text{ against } H_1: P_1 \neq P_2$$

In this case under H_0 , the test statistic,

$$\frac{(O_{12} - E_{12})^2}{(O_{12} + O_{21})} \text{ follows } \chi^2 \text{-distⁿ with 1 d.f.}$$

This test is called a McNemar's Test.

Q.4. In a department examination the candidates of both sexes yielded as presented in the following table -

Sex	Pass	Female-Fail
Male	1	5
Female	7	6

Can it be inferred that the result of the test is related to sex of the candidates?

→ Observed freq. (O_{ij}) :-

Sex	Pass	Fail	Total
Male	1	5	7
Female	7	6	13
Total	8	12	20

H_0 : The two variable are independent.

H_1 : The two variable are not independent.

$$\alpha = 0.05$$

$$\{a=1, b=5, c=7, d=6\}$$

$$N = 20$$

Expected freq. (e_{ij}) :-

$$\text{Test Statistic} = \chi^2 = \frac{\{ |ab - cd| - \frac{N}{2} \}^2}{(a+b)(c+d)(a+c)(b+d)}$$

$$= \frac{\{ | (1)(6) - (5)(7) | - \frac{20}{2} \}^2}{(1+5)(7+6)(1+7)(6+6)}$$

$$= \frac{\{ | 6 - 42 | - 10 \}^2}{(7)(13)(8)(12)}$$

$$= \frac{\{ | 1 - 36 | - 10 \}^2}{8736}$$

$$= \frac{\{ 36 - 10 \}^2}{8736}$$

$$= \frac{\{ 26 \}^2}{8736}$$

$$= \frac{676}{8736}$$

$$\chi^2 = 0.077380$$

Q.5. A sample 390 males was selected to find association between their occupation and occupation of their fathers. It revealed following information.

Father's occupation \ Son's occupation	Agriculture	Business	Service	Other
Agriculture	33	18	25	19
Business	15	29	25	30
Service	15	16	28	16
Other	17	25	31	37

Test whether the son's occupation is independent of father's occupation (use 5% level of significance)?

H_0 : Two attributes son's occupation & father's occupation

H_1 : (B) are independent

H_2 : Two attributes son's occupation & father's occupation are not independent.

Observed freq. (e_{ij}):

	B_1	B_2	B_3	B_4	Total
A_1	33	18	25	19	95
A_2	15	29	26	30	100
A_3	15	16	28	16	85
A_4	17	25	31	37	110
Total	80	88	120	102	$N = 390$

$\therefore$ Expected frequencies e_{ij} ,

$$e_{ij} = \frac{(A_i)(B_j)}{N}$$

$$e_{ij} = \frac{(A_i)(B_j)}{390} \text{ for } i, j = 1, 2, 3, 4$$

	B_1	B_2	B_3	B_4	Total
A_1	19.4871	21.4358	29.23076	24.84614	95
A_2	20.5128	22.5641	30.76923	26.15384	100
A_3	17.43589	19.17941	26.1538	24.84615	85
A_4	22.56410	24.82051	33.84616	28.76922	110
Total	80	88	120	102	390

O_{ij}^2 / e_{ij} :-

	B_1	B_2	B_3	B_4	Total
A_1	55.88289	15.114832	2132157	14.5294	106.90871
A_2	10.96875	37.27159	21.97	34.41176	104.62218
A_3	12.90441	13.347593	55.21176	11.5155	92.97934
A_4	12.80795	25.180785	28.3931	47.5855	113.9674
Total	92.56401	90.41482	26.9565	108.0423	418.47764

Test statistic is,

$$\chi^2_{(r-1)(c-1)} = \chi^2_{3 \times 3}$$

$$= \chi^2_4 = \sum_{i=1}^r \sum_{j=1}^c \left(\frac{o_{ij}^2}{e_{ij}} \right) - N$$

$$= 418.8779 - 390$$

$$\chi^2_g = 28.4779$$

Table Value,

$$\therefore \chi^2_{g, 0.05} = 16.919$$

$$\therefore \chi^2_g (\text{calculated value}) = 28.4779 > \chi^2_{g, 0.05} = 16.919$$

$\therefore$ Reject H_0 at 5% level of significance.

$\therefore$ Alternatively, P-value = $P[\chi^2_g > 28.4779] = 0.000795$
level of significance = 0.05

$\therefore$ Reject H_0

Q.c. A sample of 200 patient is drawn to test whether the audio quality of right ear is same as that of left ear.

The data obtained are follows-

Audio quality of right ear	Audio quality of left ear	
	Satisfactory	Unsatisfactory
Satisfactory	160	17
Unsatisfactory	15	08

We define attribute,

A: Quality of right ear

B: Quality of left ear

Here,

H_0 : The audio qualities of left ear & right ear are same.

H_1 : The audio qualities of right & left ears are different.

Given,

$$O_{11} = 160, O_{12} = 17, O_{21} = 15, O_{22} = 08$$

We shall apply McNeymar's test,

$$\text{Test statistic is } \chi^2_1 = \frac{(O_{12} - O_{21})^2}{O_{12} + O_{21}}$$

$$= \frac{(17 - 15)^2}{17 + 15}$$

$$= \frac{4}{32}$$

$$= \frac{1}{8}$$

$$\therefore \chi^2_1 = 0.125 \text{ [calculated value]}$$

$$\therefore \chi^2_{1, 0.05} = 3.841 \text{ [Table value]}$$

$\therefore$ Decision :- Accept H_0 at 5% level of significance.

$\therefore$ Conclusion :- The audio quality of right ear may be same as that of ear.

Q.7. A restaurant has 4 online food delivery partners. In one month, it received 1200 orders distributed as follows:

Partners	Orders
1	260
2	320
3	310
4	310

→ Test if orders are uniformly distributed?

Partner	Orders (o_{ij})	Expec. freq. (e_{ij}) [Total of o_i / Partners no. of obs.]	$(o_i - e_i)^2 / e_{ij}$
1	260	300	5.333333
2	320	300	1.333333
3	310	300	0.333333
4	310	300	0.333333
Total	1200	1200	7.333333

$$\chi^2 = 7.333333 \quad [\text{calculated value}]$$

$$\therefore D.F. (n-1) = 4-1 = 3$$

$$\therefore 7.814727903 \quad [\text{Table value}]$$

$\therefore$ Decision is-

$$7.33333 < 7.8147279$$

We fail to reject H_0 .

$\therefore$ Conclusion :-

Orders are uniformly distributed among the 4 partners.

2) Test based on t-distribution :-

Q.7. Distinguish between two sample t-test for testing equality of two population means and paired t-test?

Two Sample t-test (Independent t-test)	Paired t-test (Dependent t-test)
1. Used to compare the means of two independent populations.	1. Used to compare the means of two related or paired observations.
2. Samples are independent of each other.	2. Samples are dependent / related.
3. No relationship between observations.	3. Each observation in one sample has a corresponding pair in other sample.
4. Compare marks of class A and class B students.	4. Compare marks of same students before & after training.
5. To test whether $\mu_1 = \mu_2$ for two independent populations.	5. To test whether mean difference (μ_d) = 0.
6. Individual sample values.	6. Difference between paired values.

Q.2. A company claims that the average diameter of metal rods is 10 mm. A sample of 12 rods gives a mean diameter of 10.4 mm. The population standard deviation is known to be 1.2 mm. Test at 5% significance level whether the rods differ in mean diameter from the claimed value?

→ Given,

$$\mu_0 = 10$$

$$n = 12$$

$$\bar{x} = 10.4$$

$$\sigma = 1.2$$

$$\alpha = 0.05$$

$$\sigma^2 = \text{Pop}^n \text{ Variance}$$

$$s^2 = \text{sample variance}$$

$$H_0: \mu = \mu_0$$

$$H_1: \mu \neq \mu_0$$

$$t\text{-statistic} = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} \quad t\text{-test} = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$$

$$= \frac{10.4 - 10}{1.2 / \sqrt{12}}$$

$$= \frac{0.4}{1.2 / 3.4641}$$

$$= \frac{0.4}{0.34641}$$

$$t\text{-statistic} = 1.1547010$$

$$\therefore t\text{-Table value} = t_{\alpha/2} = t_{0.05/2} = 1.96$$

$$\therefore |t\text{-statistic}| < t\text{-Table} \rightarrow \text{We reject } H_0$$

- Confidence interval :-

- Lower C.I. & Upper C.I. :-

$$\left(\bar{x} - t_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + t_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

$$\left(10.4 - 1.96 \frac{1.2}{\sqrt{12}}, 10.4 + 1.96 \frac{1.2}{\sqrt{12}} \right)$$

$$\left(10.4 - 1.96 \frac{1.2}{3.4641}, 10.4 + 1.96 \frac{1.2}{3.4641} \right)$$

$$\left(10.4 - \frac{2.352}{3.4641}, 10.4 + \frac{2.352}{3.4641} \right)$$

$$(10.4 - 0.67896, 10.4 + 0.67896)$$

$$(9.72104, 11.07896)$$

- Conclusion :-

The rod differ in mean diameter from the claimed value.

Q.3. The ability of brakes of two new types of cars was tested by driving the cars at the

speed of 60 miles per hour and then applying the breaks the distances (in feet) required to stop the cars were noted. The results were as follows:-

Types of the cars	I	II
Number of car tested	12	15
Average distance (in inches) required to stop	7.8	9.6
S.D. of the distance	2.2	3.5

i) From above results, will you conclude that average distance required to stop using breaks of two new types of cars are equal?

ii) Find confidence Interval?

→ Let,

μ_1 : Average distance req. to stop using breaks of cars type I

μ_2 : Average distance req. to stop using breaks of cars of type II.

Here,

We want to test $H_0: \mu_1 = \mu_2$ vs

$H_1: \mu_1 \neq \mu_2$

2)

→ Test statistic :-

$$t_{n_1+n_2-2} = \frac{\bar{x} - \bar{y}}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Where,

$$s^2 = \frac{\sum (x_i - \bar{x})^2 + \sum (y_i - \bar{y})^2}{n_1 + n_2 - 2}$$

$$s^2 = \frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2 - 2}$$

Where,

S_1^2 = Variance of observations in the first Sample
= $(2.2)^2$

S_2^2 = Variance of observations in the Second Sample
= $(3.5)^2$

$$s^2 = \frac{(12)(2.2)^2 + (15)(3.5)^2}{12 + 15 - 2}$$

$$= \frac{241.83}{25}$$

$$s^2 = 9.6732$$

$$s = 3.1101768$$

$$t_{25} = \frac{7.8 - 9.6}{3.1101768 \sqrt{\frac{1}{12} + \frac{1}{15}}}$$

$$= \frac{-1.8}{1.2045659}$$

$$= -1.4943143$$

$$\therefore |t_{25}| = 1.4943143 \quad [\text{calculated value}]$$

$$\therefore t_{25, 0.05} = 2.06 \quad [\text{table value}]$$

$\therefore$ Since, H_0 is two sided the decision rule is reject H_0 if $|t_{25}|$ (calculated value) $> t_{25, 0.05}$
 2.06

$$\therefore \text{Here, } |t_{25}| = 1.4943143 \text{ (calculated value)} < t_{25, 0.05} = 2.06$$

$\therefore$ We accept H_0 at 5% level of significance.

Conclusion:-

From the given results we conclude that average distances required to stop using brakes of two types of cars are equal.

2)

$$\rightarrow \left[(\bar{x} - \bar{y}) - t^* s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}, (\bar{x} - \bar{y}) + t^* s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \right]$$

Here,

$$\alpha = 0.01, n_1 = 12, n_2 = 15, \bar{X} = 7.8, \bar{Y} = 9.6$$

$$s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} = 1.20456, t^* = t_{n_1 + n_2 - 2; \alpha/2}$$

$$= t_{25; 0.005}$$

$$= 2.787$$

$$\therefore \left[(7.8 - 9.6) - (2.787)(1.20456), (7.8 - 9.6) + (2.787)(1.20456) \right]$$

$$= [-1.8 - 3.3571, -1.8 + 3.3571]$$

$$= [-5.1571, 1.5571]$$

$$\therefore \text{Length of C.I.} = 2(3.3571) = 6.7142$$

Remark -

Our hypothesis to be tested is $H_0: \mu_1 = \mu_2 = 0$ & 0 lies in interval $[-5.1571, 1.5571]$.

So, we accept H_0 at 1% level of significance.

3) Test based on f-distribution :-

Q.1. The time taken by workers in performing a job by method I and method II is given below -

Method I :- 20, 16, 26, 27, 23, 22.

Method II :- 27, 33, 42, 35, 32, 34, 38.

Do the data show that variances of time distributions of populations from which these samples are drawn do not differ significantly?

Method I	Method II
20	27
16	33
26	42
27	35
23	32
22	34
	38

Method I X	(x - $\bar{x}$)	(x - $\bar{x}$) ²
20	-2.333	5.44289
16	-6.333	40.1068
26	3.667	13.44689
27	4.667	21.7808
23	0.667	0.4448
22	-0.333	0.11088
		81.33304

$$\bar{x} = \frac{20 + 16 + 26 + 27 + 23 + 22}{6} = \frac{134}{6} = 22.333$$

$$s^2 = \frac{\sum (x - \bar{x})^2}{n-1}$$

$$= \frac{81.33304}{6-1}$$

$$= \frac{81.33304}{5}$$

$$s^2 = 16.266608 \rightarrow \text{Variance of Method I}$$

Method II (X)	(x - $\bar{x}$)	(x - $\bar{x}$) ²
27	7.4285	55.182612
33	-1.4285	2.040612
42	7.5715	57.327612
35	0.5715	0.326612
32	-2.4285	6.044222
34	-0.4285	0.210222
38	3.5715	12.755612
Total		133.887507

$$\bar{y} = \frac{27 + 33 + 42 + 35 + 32 + 34 + 38}{7} = \frac{241}{7} = 34.4285$$

$$s^2 = \frac{\sum (x - \bar{x})^2}{n-1}$$

$$= \frac{133.887507}{7-1}$$

$$= \frac{133.887507}{6}$$

$$s^2 = 22.28571 \rightarrow \text{Variance of Method II}$$

F statistic

$$F\text{-statistic} = \frac{\text{Large Variance of Method I \& II}}{\text{Small Variance of Method I \& II}}$$

$$= \frac{22.28571}{16.266608}$$

$$F\text{-statistic} = 1.370028$$

Table value :-

$$F_{\alpha/2, df_1, df_2}$$

$$df_1 = 5$$

$$df_2 = 6$$

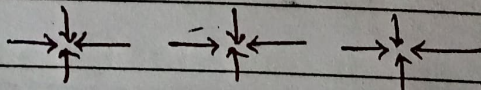
$$\alpha = 0.05$$

$$\therefore F_{\text{Table}} = 3.404507$$

$$\therefore 1.370028 < 3.404507 \rightarrow \text{Accept } H_0.$$

- ~~Conclusion :-~~

~~These samples are drawn do not differ significantly.~~



~~By the~~
17/02/2028
V. good.